

Knights of the Old Republic High-Resolution Menu Guide

There are two things necessary to get your KotOR game menus functioning at high resolution:

1. A modified `swkotor.exe` file, customized for your resolution.
2. A set of GUI files, also customized for your resolution, placed in your `SWKotOR/Override` directory.

This guide assumes you have *already* used something like UniWS to get basic game support for your resolution. This guide *only* covers scaling the main menu, game menus, and load screens.

Note on Letterbox Scaling

The executable patcher *can* adjust the scale of the dialog letterbox interface to match the original game's proportions. This is only useful for **non-4:3** screen resolutions. In fact, if you ask the patcher to use this feature for a 4:3 resolution, it will ignore you. Using this fix will show *less* of the action in the middle--some people will not like that, so the behavior is **optional**. If you are using a super-widescreen resolution such as 21:9, the game's dialog is not visible at all without this fix.

Installation / Getting this to work properly

Step 0: Find the GUI folder matching your target resolution

Find the `gui.WIDTHxHEIGHT` folder in this package matching the WIDTH and HEIGHT for the resolution you want to use. They are grouped by aspect ratios, so 1920x1080 is at `16-by-9/gui.1920x1080`. **If you do not find the WIDTH and HEIGHT you need, this package is unlikely to produce useable results.** You may have a currently unsupported resolution.

Step 1: Modified Executable `swkotor.exe`

Modifying a binary executable requires a hex editor capable of modifying specific bytes (via their hexadecimal values). The only other option is a script or program capable of making the modifications for you. Use *one* of the following two techniques.

Technique 1: Automatic Modification with Patching Script

The High-Resolution Menu package includes a patching script. The current version is very limited in behavior and will likely only work with a GOG or 4-CD executable, or any executable compatible with UniWS.

Windows Instructions

Assuming you want to patch C:\GOG Games\SWKotOR\swkotor.exe to 1920x1080 resolution and you *do* want to adjust your letterbox proportions to match the original 4:3 proportions:

1. Either copy hires_patcher.bat & hires_patcher.exe to your SWKotOR game directory, OR, copy swkotor.exe to the directory containing hires_patcher.bat & hires_patcher.exe. This step is not necessary but it makes things easier.
2. Double-click hires_patcher.bat.
3. Provide the 4 pieces of information as instructed, in this case: `1920` , `1080` , `yes` , `swkotor.exe` .
4. Copy the patched exe (and backup) to your game directory, if necessary.

The patcher, if successful, would create a backup of the original executable file, `swkotor.bak.exe` .

If the patcher succeeds, you should see output like this:

```
STEP 1: Tests passed, all patch offsets found, proceeding...
STEP 2: Backup file created: C:\GOG Games\SWKotOR\swkotor.bak.exe
STEP 3: Patching executable: C:\GOG Games\SWKotOR\swkotor.exe
Patch successful
```

If the patcher fails, you should see some kind of error(s) followed by

```
ERROR: no modifications made, patch unsuccessful
```

If you provided invalid data, you would see some kind of error(s) followed by the basic command line help:

```
Patch Knights of the Republic swkotor.exe for proper operation with rescaled UI.

usage: hires_patcher.exe WIDTH HEIGHT LETTERBOX_SCALE EXECUTABLE

WIDTH is the horizontal resolution
HEIGHT is the vertical resolution
LETTERBOX_SCALE fix proportions yes | no (or true | false, 1 | 0)
EXECUTABLE is the path to a writeable swkotor.exe file
```

MacOS / Linux Instructions

In order to attempt this option, you must be able to use the command line. Use the hires_patcher.pl tool like this (your file paths will be different):

```
./hires_patcher.pl 1920 1080 yes /home/user/drive_c/Program Files/SWKotOR/swkotor.exe
```

Where `/home/user/drive_c/Program Files/SWKotOR/swkotor.exe` is a path to the swkotor.exe file you wish to patch.

MacOS Alternative Instructions

The patcher can also modify some "mac-native" x86 versions of Knights of the Old Republic.app. In order to do this, follow the MacOS / Linux Instructions, but replace the path to the .exe file with the path to the XKOTOR binary within your app (use "Show Package Contents" to find this file). For example:

```
./hires_patcher.pl 1920 1080 yes /Applications/Knights\ of\ the\ Old\ Republic.app/Contents/MacOS/XKOTOR
```

You will also have to move the backed up XKOTOR.bak file away from Contents/MacOS if the patcher completes successfully.

Technique 2: Manual Modification with a Hex Editor

- Prepare your hex editor, make sure you understand how to use it.
- You need to figure out what your resolution (X by Y) is in 16-bit little-endian hexadecimal, in *both* positive and negative values. So 4 total numeric/hexadecimal values (+X, +Y, -X, -Y). For your convenience, I am providing this table of common screen resolution values:

Numeric	Positive Hex	Negative Hex
720	D0 02	30 FD
768	00 03	00 FD
800	20 03	E0 FC
900	84 03	7C FC
1024	00 04	00 FC
1050	1A 04	E6 FB
1080	38 04	C8 FB
1152	80 04	80 FB
1200	B0 04	50 FB
1280	00 05	00 FB
1366	50 05	B0 FA
1440	A0 05	60 FA
1600	40 06	C0 F9
1680	90 06	70 F9
1920	80 07	80 F8
2140	5C 08	A4 F7
2560	00 0A	00 F6
3840	00 0F	00 F1

- If you wish to attempt the letterbox scaling modification, you need to make the following calculations, convert your result to a 4-byte floating point number, then invert the order of the bytes (every 2 hex characters). This is your 'letterbox scaling value'. Calculation: $1.0 / ((\text{width} / \text{height}) * 1.75)$ Example: $1.0 / ((1920 / 1080) * 1.75) = 0.3214286 =$ 3E A4 92 4A => 4A 92 A4 3E
- In this example, we will use **1920 x 1080** as our target resolution. This means that +X = 80 07 , -X = 80 F8 , +Y = 38 04 , -Y = C8 FB . The following example offsets are for the GOG edition of the game, if you have a different version, offsets may differ.
- **Make a backup of your swkotor.exe before you load it in your hex editor!!!**

- Load swkotor.exe in your hex editor.
- You must make 3 sets of 2 edits, for a total of 6 changed values (12 changed bytes).
- First edit: this will fix being able to click menu buttons

Around offset B6C0h (46784), find bytes that look like this:

```
78 00 8B 4C 24 24 05 80 FD FF FF 99 2B C2 D1 F8
F7 D8 03 C8 A1 D8 D1 78 00 05 20 FE FF FF 99 2B
```

In the above, you need to replace `80 FD` with your -X value, and `20 FE` with your -Y value, so, with our example values (`80 F8` & `C8 FB`), *after* editing, it would look like this:

```
78 00 8B 4C 24 24 05 80 F8 FF FF 99 2B C2 D1 F8
F7 D8 03 C8 A1 D8 D1 78 00 05 C8 FB FF FF 99 2B
```

- Second edit: this will fix being able to scroll and click list items (like save games)

Around offset BA60h (47712), find bytes that look like this:

```
89 54 24 18 75 3C A1 D4 D1 78 00 05 80 FD FF FF
99 2B C2 8B 16 D1 F8 F7 D8 03 D0 89 16 A1 D8 D1
78 00 05 20 FE FF FF 99 2B C2 8B 17 D1 F8 F7 D8
```

In the above, you need to replace `80 FD` with your -X value, and `20 FE` with your -Y value, so, with our example values (`80 F8` & `C8 FB`), *after* editing, it would look like this:

```
89 54 24 18 75 3C A1 D4 D1 78 00 05 80 F8 FF FF
99 2B C2 8B 16 D1 F8 F7 D8 03 D0 89 16 A1 D8 D1
78 00 05 C8 FB FF FF 99 2B C2 8B 17 D1 F8 F7 D8
```

- Third edit: this will fix the position of load screens

Around offset AA60h (43616), find bytes that look like this:

```
0F BF 42 6C 2D 80 02 00 00 99 2B C2 8B 16 D1 F8
03 D0 89 16 F6 41 44 40 74 19 8B 41 18 0F BF 40
6E 8B 4E 04 2D E0 01 00 00 99 2B C2 D1 F8 03 C8
```

In the above, you need to replace `80 02` with your +X value, and `E0 01` with your +Y value, so, with our example values (`80 07` & `38 04`), *after* editing, it would look like this:

```
0F BF 42 6C 2D 80 07 00 00 99 2B C2 8B 16 D1 F8
03 D0 89 16 F6 41 44 40 74 19 8B 41 18 0F BF 40
```

```
6E 8B 4E 04 2D 38 04 00 00 99 2B C2 D1 F8 03 C8
```

- Fourth edit (**optional**, letterbox scaling): this will change the proportions of the dialog letterbox interface

Around offset 355788h (3495816), find bytes that look like this:

```
64 75 6D 6D 79 00 00 00 64 75 72 61 74 69 6F 6E
00 00 00 00 00 00 7A C4 B9 6D DB 3E 00 00 00 00
D0 8A 6A 00 A0 A5 40 00 30 A6 40 00 60 E7 60 00
```

In the above, you need to replace `B9 6D DB 3E` with your new letterbox scaling value, so, with our example value (`4A 92 A4 3E`), *after* editing, it would look like this:

```
64 75 6D 6D 79 00 00 00 64 75 72 61 74 69 6F 6E
00 00 00 00 00 00 7A C4 4A 92 A4 3E 00 00 00 00
D0 8A 6A 00 A0 A5 40 00 30 A6 40 00 60 E7 60 00
```

- **Because you made a backup copy before you started**, it is now safe to save these changes, so do that.

Step 2: Copy GUI files to Override folder

Find the folder in the High-Resolution Menu package that matches the resolution you patched your executable for in Step 1.

Copy the .gui files from the chosen directory into the Override folder for your game.

Following the example in Step 1, if you patched using 1920x1080 in Step 1, copy the files from

```
gui.1920x1080 into C:\GOG Games\SWKotOR\Override
```

Uninstall

Restore your backed up executable, remove the gui files from your Override

Known Issues

- The patcher is currently incredibly limited, making no effort to search for values, only looking at specific locations where they are in the executables I have access to. It won't work for the Steam executable.
- This package does not contain any higher resolution artwork. At high resolutions, menu backgrounds will be blocky, and the striped list backgrounds are unlikely to line up with list items. For numerous reasons,

you will want to get a high resolution menu art package.

- The game's initial character selection screen does not lay out the selectable characters properly according to GUI file field values. It is useable, but not perfect. The blue hilight borders had to be removed from around the characters to maintain a clean look.
- The map screen is scaled up, but the map itself is not. Furthermore, the position of the map within the frame is somewhat random/poorly understood. The map, however, is completely useable, and the points of interest and player indicator are all positioned accurately relative to the map.
- Some of the 3D models in menus, such as on the character stats or character generation screens, display line artifacts over them. This is caused by the very low quality graphics being scaled up to provide the border overlays. Simply extracting the graphics as TGAs, scaling them up, and placing them into Override/ sometimes solves these problems.
- Rescaling the letterbox seems to introduce occasional issues, particularly when the letterbox animation happens. This may be fixed at some point in the future.